TRIBHUVAN UNIVERSITY

INSTITUTE OF SCIENCE AND TECHNOLOGY

A FINAL YEAR INTERNSHIP REPORT ON

" QUALITY ASSURANCE/QUALITY CONTROL AT ALPINIST STUDIOS PVT. LTD."

In partial fulfillment of the requirements for the Bachelor's Degree in Information Technology (BIT)

Under the Supervision of

Mr. Ramesh Singh Saud

Lecturer

Padmakanya Multiple Campus

Submitted By:

Sabu Bhuju

TU Exam Roll No.: BIT 535/078

TU Registration No.: 5-2-38-339-2021

Submitted To:

Padmankanya Multiple Campus

Department of Information Technology

Bagbazar, Kathmandu

April, 2026

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MENTOR'S RECOMMENDATION

Alpinist Studios

VAT No. : 604263997
Regd No. 149747/72/73

Date: 9th March 2026

To whom it may concern

I am writing to recommend **Ms. Sabu Bhuju**, who completed a **3-month internship** as a **Quality Assurance (QA) & Quality Control (QC)** at **Alpinist Studios** from **December 1, 2025 to February 28, 2026.** I had the opportunity to mentor her during this period.

Ms. Bhuju was willing to learn and open to feedback throughout her internship. She completed her tasks responsibly, asked questions when needed, and showed steady improvement over time. She made an effort to understand the purpose behind QA processes, not just follow them mechanically.

During her internship, she developed practical skills in software testing and quality control under my guidance and that of our senior team members. She contributed to test planning, bug reporting, and verification tasks, and grew more familiar with identifying issues and communicating them clearly to the development team.

I recommend Ms. Bhuju for roles in QA and QC. She has built a solid foundation during her internship and has the right mindset for continued growth in this field.

I wish her success in her career.

Thank You.

Regards,

Alpinist Studios

.....
Ghanashyam Kushwaha
Sr. Tech Project Manager

Tapalhon, Manbhawan Rd
Lalitpur, Nepal

(+977-1) 5432671

www.alpiniststudios.com
info@alpiniststudios.com

CORPORATE CERTIFICATE

Alpinist Studios

VAT No. : 604263997
Regd No. 149747/72/73

Date: 9th March 2026

To whom it may concern

This is to certify that **Ms. Sabu Bhuju** successfully completed her **3-month internship** at **Alpinist Studios** as a **Quality Assurance (QA) & Quality Control (QC)** Intern, from **December 1, 2025 to February 28, 2026.**

Ms. Bhuju was dedicated, a quick learner, and a hardworking member of our team throughout her internship. She followed instructions diligently and made commendable progress on her assigned tasks. With the guidance of her mentor, she demonstrated great potential for a successful career in her field.

Ms. Bhuju's contributions were significant, and she proved to be an invaluable part of our team.

We wish her the very best in her future endeavors.

Thank You.

Regards,

.....

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SUPERVISOR'S RECOMMENDATION

This is to recommend that **SABU BHUJU** Exam Roll No.: **BIT 535/078**, T.U. Registration No.: **5-2-38-339-2021** has carried out an internship report entitled **'Quality Assurance and Quality Control at Alpinist Studios Pvt. Ltd."** for partial fulfillment of the requirements for the degree Bachelor in Information Technology under my supervision. To my knowledge, this work has not been submitted for any degree.

She has fulfilled all the requirements laid down by the Padmakanya Multiple Campus. Department of Information Technology, Bagbazar, Kathmandu.

Mr. Ramesh Singh Saud
Internal Supervisor
Padmankanya Multiple Campus
Department of Information Technology
Bagbazar, Kathmandu

Date: April 2026

LETTER OF APPROVAL

This is to verify that this internship report prepared by Sabu Bhuju entitled **"Quality Assurance and Quality Control at Alpinist Studios Pvt. Ltd."** in partial fulfillment of the requirements for the degree of Bachelor in Information Technology has been well studied. In our opinion, it is satisfactory in scope and quality as required for the degree.

Evaluation Committee

Mr. Ramesh Singh Saud
(Internship Supervisor)

Mrs. Sunita Shrestha
(Program Coordinator)

External Examiner
Tribhuwan University

DECLARATION

I hereby declare that the internship report entitled **'Quality Assurance and Quality Control at Alpinist Studios Pvt. Ltd.'**, submitted to the Department of Information Technology, Padmakanya Multiple Campus is my original work. This project has not been submitted earlier in part or full in this or any other form to any university or institute, here or elsewhere, for the award of any degree. This work has been carried out in this form in partial fulfillment of the requirements of the Bachelor's Degree in Information Technology (BIT) under the guidance of **Mr. Ramesh Singh Saud.**

Sabu Bhuju

TU Exam Roll No.: BIT 535/078

TU Registration No.: 5-2-38-339-2021

ACKNOWLEDGEMENT

The successful completion of their internship report would not have been possible without the support and assistance of many individuals and organizations. I feel immensely blessed to have gotten during the of my internship program. I would like to take the opportunity to offer my earnest admiration to each one of them.

First and foremost, I am highly indebted to **Mr. Ashok Maharjan** who took confidence in me and provided me with the opportunity to work as an Intern at **Alpinist Studios Pvt. Ltd.** I had a wonderful and an unforgettable experience being part of such a lovely and lively team.

I express my sentiment of gratitude to **Mr. Ghanshyam Kushwaha** , a Senior Tech Project Manager, who has been a continuous lead of inspiration as my intern mentor. Without his constant guidance and suggestions, the report would have been nowhere near completion. My gratitude for her trust and generosity goes beyond words.

My special thanks to **Mr. Ramesh Singh Saud** for his constant guidance towards preparing their report standard as per the norms and values.

Finally, my thanks and appreciations go to each one of my colleagues, and the entire Alpinist Studios team, who irrespective of the situation, always engaged and supported me to prepare the report.

Thank you,
Sabu Bhuju
TU Exam Roll No.: BIT 535/078
TU Registration No.: 5-2-38-339-2021

ABSTRACT

This report presents the experience and learning outcomes gained during the internship as a Quality Assurance and Quality Control (QA/QC) Intern at Alpinist Studios. The main objective of the internship was to gain practical knowledge of software testing processes and understand how quality assurance contributes to the development of reliable and high-quality software applications.

During the internship, the intern was primarily involved in testing web and mobile applications. The responsibilities included preparing and executing test cases, identifying defects in the system, reporting bugs, and validating application functionality. The testing activities mainly focused on functional testing, UI validation, responsive and cross-browser testing, and API testing to ensure that the applications met the required specifications and provided a smooth user experience.

In addition to manual and API testing, the internship also provided exposure to automation testing using Playwright, along with tools such as Postman, Swagger, and Jira for API validation and bug tracking. The intern learned how automation testing supports regression testing and improves efficiency and accuracy in the software testing process.

Overall, the internship provided practical exposure to real-world software testing practices and helped develop analytical, problem-solving, and documentation skills related to quality assurance in web and mobile applications.

Keywords: *Quality Assurance (QA), Manual Testing, Automation Testing, Playwright, API Testing, Postman, JIRA, Software Testing Life Cycle (STLC)*

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LIST OF ABBREVIATIONS

API	Application Programming Interface
APK	Android Package Kit
CRUD	Create, Read, Update, Delete
HLDC	Himalayan Leadership Development Center
HR	Human Reaource
HRMS	Human Resource Management System
HTTP	HyperText Transfer Protocol
IT	Information Technology
MVP	Minimun Viable Product
OTP	One-Time Password
QA	Quality Assurance
QC	Quality Control
SDLC	Software Development Life Cycle
STLC	Software Testing Life Cycle
UI/UX	User Interface / User Experience
URL	Uniform Resource Locator

CHAPTER 1: INTRODUCTION

1.1 Introduction

This report titled "Quality Assurance and Quality Control at Alpinist Studios" aims to give a comprehensive overview of the tasks undertaken, the learning experiences gained during the internship period as a Quality Assurance and Quality Control (QA/QC) Intern. This internship provided an opportunity to learn about the practical application of software testing methodologies in a professional software development environment. The main aim of the internship was to ensure the quality, reliability, and functionality of software applications through a series of testing and validation processes.

The role as a QA/QC intern involved carrying out various activities to ensure the quality of the software application. These activities included carrying out manual testing, API testing, and automation testing to ensure the application meets the requirements by detecting defects. Manual testing activities included analyzing the application's features, creating test cases, executing test cases, recording the test results, and reporting the bugs detected during the testing process. These activities ensured the application's functionality to provide a reliable and efficient user experience.

During my internship, active contributions were made to the testing of several software products, including a **Human Resource Management (HRM) system**, the **SeekNFind** mobile application, and the **HLDC Nepal** website. Responsibilities were mainly focused on ensuring that these systems functioned correctly, identifying defects, and verifying that the applications met the required functional and usability standards.

During the internship period, practical knowledge about the Software Testing Life Cycle (STLC) and how to report bugs effectively as a team was obtained. This report describes the activities performed during the internship period, techniques used during the internship period, and the learning experience obtained during the internship period, with a particular emphasis on the importance of quality assurance in developing efficient software systems.

The internship period commenced in 01 December, 2025 and ended in 28 February, 2026.

1.2 Problem Statement

The current environment depends on technology which has become vital for current society, and software applications are essential for all business processes. Web and mobile applications function as the primary platforms through which organizations provide services, handle data, and create better user experiences. The applications must meet quality standards because users want to experience uninterrupted performance across all their devices and platforms.

The HLDC Website experienced performance problems because the Figma design did not match the user interface that was built. Users experienced problems with blog pagination, post loading, and category filtering, which affected the website's usability and overall user experience.

The SeekNFind Mobile Application encountered challenges with user input validation and profile management and photo handling. The operational difficulties emerged because uploaded photos failed to display in the app and deleted photos vanished from the app while staying on the server and inconsistent API responses occurred. The app faced decreased reliability because of these issues which affected its complete operational capacity.

The Human Resource Management (HRM) System encountered problems when trying to combine its employee management system with its leave request system and its attendance tracking system. The system experienced API failures which prevented proper operation and completed offboarding activities without verifying checklists, which resulted in offboarded employees maintaining system access. The system exhibited input validation problems which indicated a requirement for more stringent data protection and access restriction systems.

1.3 Objectives

The internship was undertaken to provide students with a comprehensive learning experience by integrating academic knowledge with practical industry application. The main objective are:

1. To gain practical knowledge of software testing processes and Quality Assurance (QA) practices in a professional development environment.
2. To perform manual testing, automation testing, and API testing to verify application functionality and identify defects.
3. To learn test case creation, defect reporting, and bug tracking to ensure software quality and reliability.
4. To develop professional skills such as teamwork, communication, and problem solving while working in a software development team.

1.4 Scope and Limitations

1.4.1 Scope

The scope of this internship mainly focused on learning and applying software testing techniques in real-world projects at Alpinist Studios. The activities included manual testing, API testing, and user interface testing of web and mobile applications.

The internship involved testing the **HRM system**, the **SeekNFind mobile application**, and the **HDLC website**. Tasks included analyzing system requirements, preparing test cases, executing tests, validating API responses, and reporting defects to ensure the proper functionality and usability of the applications.

The internship also provided exposure to software testing concepts such as testing methodologies, defect management, Agile development practices, and basic automation and performance testing concepts.

1.4.2 Limitations

1. The internship period was limited, which restricted deeper involvement in advanced automation testing implementation.
2. Most of the testing activities were focused on manual and API testing, while automation testing exposure was limited to introductory concepts.
3. Access to some internal systems and production environments was restricted due to company security policies.
4. The testing activities were limited to specific company projects and may not represent all real-world software testing scenarios.

1.5 Report Organization

This report primarily highlights the work performed and learning experiences gained during the Quality Assurance and Quality Contol(QA/QC) internship at Alpinist Studios. The report is organized into four main chapters, each focusing on different aspects of the internship and the skills developed throughout the internship period.

Chapter 1: Introduction

This chapter presents an overview of the internship report, including the background and purpose of the internship. It outlines the problem statement, objectives, scope, and limitations of the internship. The chapter also introduces the importance of quality assurance in software development and explains how the internship helped bridge the gap between academic knowledge and real-world QA practices.

Chapter 2: Organization Details and Literature Review

This chapter provides detailed information about the organization, Alpinist Studios, including its introduction, organizational structure, working domains, and the department where the internship was conducted. It also includes a literature review covering quality assurance concepts, software testing life cycle (STLC), testing methodologies, and tools that formed the theoretical foundation for the internship work.

Chapter 3: Internship Activities

This chapter describes the roles and responsibilities carried out during the internship. It includes weekly logs of tasks performed, such as manual testing, test case preparation, bug reporting, API testing using Postman, automation testing using Playwright with JavaScript, and test management using Jira. The chapter also explains the projects involved during internship period, tools used, challenges faced, and technical skills gained during the internship.

Chapter 4: Conclusion and Learning Outcomes

This chapter summarizes the overall experience, key achievements, and learning outcomes of the internship. It highlights the practical skills developed in areas such as manual testing, API testing, automation testing, STLC, SDLC, and defect tracking. The

chapter also reflects on professional growth, teamwork experience, and how the internship contributed to future career preparation in the field of quality assurance.

The report also includes references and appendices containing supporting materials such as screenshots, code snippets.

CHAPTER 2: ORGANIZATION DETAILS AND LITERATURE REVIEW

2.1 Introduction to Organization

Alpinist Studios Pvt. Ltd., formally known as **SoAni Tech Pvt. Ltd.** is well-known as a digital engineering company established in 2015 that specializes in the development of software products, the implementation of technology solutions, and the provision of dedicated IT services to businesses worldwide.

With the aim of turning concepts into reality, Alpinist Studios collaborates with a global team designed to provide high-quality solutions for clients in various sectors, including Healthcare, Education, Non-profits, and Retail.

The firm offers a wide range of services, including MVP development, staff augmentation, external team deployment, and IT consulting. Alpinist Studios excels in creating mobile, web, and custom applications, catering to both intricate enterprise systems and initial-stage MVPs, with a strong commitment to assisting startups in realizing their innovative concepts.

Figure 1: Organization Logo of Alpinist Studios

For companies aiming to expand their teams, Alpinist Studios supplies skilled and experienced IT professionals on either a temporary or permanent basis, aligning their team with the client's for smooth product development. When it comes to consulting Alpinist Studios does a lot of work to help people with their computer systems. They make sure that each client gets the help they need to do in the world of computers, which is always changing.

Alpinist Studios uses the methods to develop software. They work closely with their teams and clients to make sure that the things they make are good and work well. Alpinist

Studios has worked with some names, like the Government of Nepal, Ghana Water Company Limited, LG, Timeero and a lot of other companies.

Table 1: Organization Details

Organization	Alpinist Studios Pvt. Ltd.
Telephone	01-5432671
Email	info@alpiniststudios.com
Official Website	https://alpiniststudios.com/
Location	Manbhawan Rd, Lalitpur, Nepal
Opening Hour	9:00 A.M. - 5:30 P.M.

2.2 Organizational Hierarchy

Alpinist Studios operates through a well-defined organizational hierarchy which enables the company to execute projects and deliver results without disruption. The **Chief Executive Officer (CEO)** functions as the highest-ranking official who establishes overall company direction through his leadership and strategic decision-making.

The **HR/Administration Manager**, the **Business Analyst,** and the **Accounts Department** function as essential components that enable the organization to operate efficiently which reports directly to the CEO.

The **Project Manager** operates as the main driver of daily activities by establishing workflow patterns while overseeing project schedules and maintaining open lines of communication between all teams.

The studio's technical infrastructure consists of highly competent professionals who include **Software Developers** (Frontend and Backend) and **UI/UX Designers** and **DevOps Engineers** and **Quality Assurance (QA) Engineers** who provides assistance to interns. The product development process involves these teams working together from the initial design stage until the final testing and deployment phase to create software solutions that meet client needs.

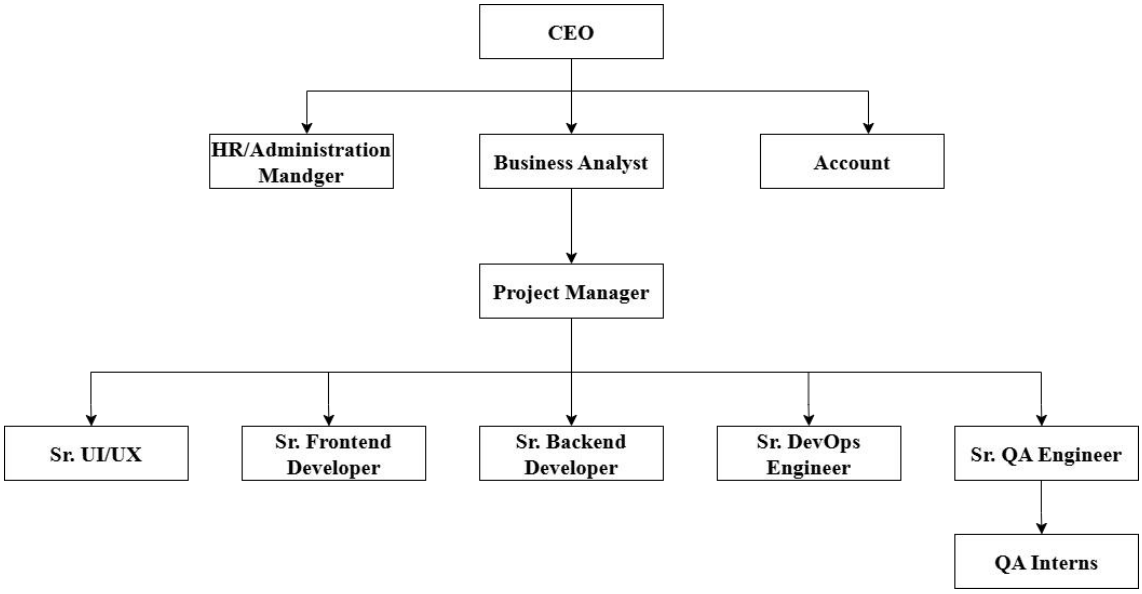


Figure 2: Organizational Hierarchy of Alpinist Studios Pvt. Ltd.

2.3 Working Domains of Organization

Alpinist Studios provides various technology services which fulfill the ongoing requirements of businesses in all market sectors. Their core service offerings include:

1. Web Application Development:

Alpinist Studios designs and develops scalable secure and high-performance web applications which meet the specific business needs of each client to achieve dependable operational performance and future system upkeep.

2. **Mobile Application Development:**

The studio builds both cross-platform and native mobile applications for Android and iOS with a strong emphasis on enhancing user accessibility engagement and overall experience across devices.

3. MVP Development:

Alpinist Studios supports startups and enterprises alike in bringing their ideas to life through Minimum Viable Product MVP development. This service helps clients validate concepts assess market feasibility and accelerate their path to product launch with reduced risk.

4. API Development and Integration:

The technical team creates secure APIs which enable system interoperability and third-party software connections and microservices-based digital system operations.

5. UI/UX Design:

Alpinist Studios design team develops user interfaces which prioritize user experience following modern design standards and accessibility requirements. The team aims to create experiences which combine visual attractiveness with user-friendly functionality to assist end users.

6. Dedicated Development Teams and Staff Augmentation:

For businesses looking to scale their technical capacity Alpinist Studios provides dedicated external development teams and skilled IT professionals who integrate seamlessly with client workflows.

2.4 Description of Intern Department/Unit

The internship program at Alpinist Studios functions through its established framework which determines specific duties assigned to each intern. Interns receive department assignments which match their abilities and educational qualifications and their preferred career paths. Frontend and backend interns work under the guidance of their respective development mentors, UI/UX interns receive assistance from the design team, and QA interns work under the tech project manager who manages all testing operations to uphold quality standards during development.

Interns receive daily assignments which match actual project needs and their work performance gets assessed through three criteria which include their task execution and product quality and their overall development. Alpinist Studios creates a friendly and professional work environment which helps interns develop their software development skills at a comfortable pace.

The Tech Project Manager who controlled the QA Intern work provided access to the web application which operated in a localhost environment to test particular features. The team developed test cases which they executed to test all application features while using

Jira and Google Sheets to report software defects. The Project Manager reviewed all reported defects to check their validity before developers received them for correction. The team conducted retesting after implementing fixes to confirm that all identified problems had been fully resolved.

This internship provided essential practical experience in performing manual testing while designing test cases and tracking defects and working with multiple teams in a real software development workplace because it helped to learn about the Software Testing Life Cycle (STLC) and standard industry quality assurance procedures.

Table 2: Internship Details

Start Date	01 December, 2026
End Date	28 February, 2026
Working Hours	8.5 hours per day (9:00 A.M. - 5:30 P.M.)
Working Days	5 days a week (Monday - Friday)
Mentor's Name	Ganashyam Kushwaha
Position	QA/QC Intern
Internship Period	12 weeks

2.5 Literature Review / Related Study

Modern software development needs Quality Assurance (QA) and Quality Control (QC) methods to combine their efforts in creating high-quality software that can be developed in quickly changing situations. Continuous integration and continuous testing maintain software quality because improved QA visibility and effective collaboration between developers and testers and optimized documentation all contribute to better development processes (Ladokun et al., 2024). QA combines quality practices with all development stages to prevent defects while QC uses testing to find defects in the completed product. The combination of these methods with automated testing systems results in enhanced operational efficiency and dependable performance which leads to superior product quality that enables organizations to deliver software products which fulfill both high industry standards and user requirements.

Manual testing operates as the primary testing technique for software evaluation because it requires testers to create and run test cases through manual testing without using any automated testing tools. The method evaluates software performance to see whether it meets user requirements yet takes extended time periods for testing (Kumari & Chauhan, 2018). Automated testing, in contrast, relies on scripts and software to perform tests automatically, which enables testing to operate at higher speed and better reliability and greater precision while saving testing time for human testers (Gamido & Gamido, 2019). The modern testing frameworks which Playwright provides work together with major web browsers to support testing of multiple programming languages while their automatic waiting system and concurrent testing capability and separate testing environment system. The testing system enables testers to reproduce actual user behavior which lets them evaluate application performance while testing becomes more efficient and software quality improves (Microsoft, n.d.).

API testing guarantees proper operation of software systems while maintaining their expected performance and security standards within complex SAP systems. The testing processes become more efficient through the combined functionality of Postman and Swagger together with their structured testing system which supports testing activities (Westerveld, 2024). Postman provides testers with essential tools to build HTTP requests execute authentication tasks develop automated testing procedures and establish links to

CI/CD platforms (Perabathini, 2019). Swagger uses its OpenAPI Specification to provide interactive API documentation which enables users to explore endpoints and create client libraries through automated processes (Perabathini, 2019). Postman focuses on executing tests and automating processes while Swagger serves as a tool for creating APIs and documenting their functionality and testing their endpoint operations.

The software development process relies on bug tracking systems to provide teams with tools for defect identification and documentation and defect management and project progress tracking. The Jira system is favored by many teams due to its capability to document and disseminate bugs, while also utilizing automated workflows to manage their development tasks (Atlassian, n.d.). The integration of Zephyr Squad with the system provides testing capabilities that enable users to develop and arrange test cases, perform test executions, and connect requirements with defects to establish full traceability (Fudge, 2023). The automation testing integrations of Jira work together with its reporting and collaboration tools to help teams track project development while they detect problems and distribute project information. The combination of these functions enables Jira to deliver software development processes that result in high-quality and dependable outcomes.

CHAPTER 3: INTERNSHIP ACTIVITIES

3.1 Roles and Responsibilities

The position of QA Intern at Alpinist Studios necessitates my engagement in various tasks which involve assessing software applications through operational efficiency tests and reliability performance tests and through quality assessment tests. The role entails the execution of the following responsibilities.

1. To develop test cases and scenarios derived from software specifications and application features.
2. Executing manual testing on both web and mobile platforms to uncover defects and verify correct functionality.
3. To perform API testing utilizing Postman for the validation of backend interactions.
4. Evaluating React-based web applications to guarantee UI uniformity, responsiveness, and user-friendliness.
5. Assessing Flutter mobile applications on both Android and iOS systems.
6. To document and monitor defects through Jira and Google Sheets.
7. Working collaboratively with the team to replicate issues and assist in bug resolution.
8. Conducting retesting and regression testing following fixes to ensure system stability and identify any new defects.

Through these responsibilities I gained practical experience with the Software Testing Life Cycle (STLC) which I used to understand how software development organizations perform quality assurance testing.

3.2 Weekly Log

Table 3: Weekly Log

Weeks	Logs/ Activities
Week 1 [Dec 01 to Dec 05]	• Attended company orientation and completed onboarding and understood the QA workflow and team structure. • Learned about basic manual testing including test planning, test case development and bug reporting. • Set up work accounts: Slack, Jira, Postman, and company email.
Week 2 [Dec 08 to Dec 12]	• Got introduced to all three assigned projects: HLDC Website, SeekNFind, and HRM System. • Received access to HLDC Website and reviewed all pages against Figma design reference. • Performed UI testing across all pages; verified layout, fonts, spacing, colors, and images. • Checked responsiveness on desktop, tablet, and mobile using browser DevTools. • Reported all bugs using Google Sheets with annotated screenshots.
Week 3 [Dec 15 to Dec 19]	• Performed functional testing on HLDC Website contact form, navigation links. • Tested contact form field validations, submission success message, and error handling. • Verified blog section pagination, post loading, and category filtering behavior. • Checked cross-browser compatibility on Chrome, Firefox, and Edge. • Performed regression testing after developer fixes and confirmed previously found UI bugs are resolved.

Week 4 **[Dec 22 to Dec 26]**	• Reviewed SeekNFind project requirements and user stories with Project Manager. • Installed the APK on an Android device and manually explored all application screens. • Compared app screens against Figma design to understand expected behavior. • Studied user stories documented in Confluence via Jira to understand the functional requirements and expected system behavior.
Week 5 **[Dec 29 to Jan 02]**	• Set up Postman collection for SeekNFind APIs and configured environment variables for base URL and auth tokens. • Tested login, logout, and token refresh endpoints. • Sent requests with missing, expired, and invalid tokens to verify error handling. • Tested POST endpoints for creating listings and verified data was saved correctly.
Week 6 **[Jan 05 to Jan 09]**	• Continued API testing for SeekNFind; tested church search and listing GET endpoints and validated response structure, data accuracy, and status codes. • Tested member management module verified adding, viewing, and removing members from a church. • Verified that member roles and permissions were correctly assigned and reflected in the application. • Tested event management module created, edited, and deleted events and verified correct data display • Performed regression testing on previously fixed bugs and organized Postman collection by module.
Week 7 **[Jan 12 to Jan 16]**	• Wrote test cases for the Registration and Login modules in Google Sheets based on the reviewed user stories. • Executed test cases for Registration, Login and Profile modules; tested valid and invalid credential scenarios, empty

	fields, and duplicate account registration. ● Verified OTP verification flow, session persistence, and profile image upload functionality. ● Retested previously fixed bugs from the earlier build and confirmed resolution.
Week 8 **[Jan 19 to Jan 23]**	● Received access to HRM System and reviewed modules employee management, attendance, leave, and payroll. ● Set up Postman collection for HRM APIs and tested employee CRUD endpoints. ● Verified role-based access by testing endpoints with admin and regular user tokens. ● Discussed HRM testing scope and priorities with Tech Lead for upcoming weeks.
Week 9 **[Jan 26 to Jan 30]**	● Performed manual testing of HRM modules employee forms, leave requests, and attendance tracking. ● Tested form validations, date pickers, dropdowns, and mandatory field checks. ● Got introduced to Playwright automation framework by Tech Lead. ● Read Playwright documentation and ran existing test scripts to understand test structure.
Week 10 **[Feb 02 to Feb 06]**	● Wrote automated test scripts for HRM Login module using Playwright with JavaScript. ● Automated valid login, invalid credentials, empty field, and session handling test cases. ● Used Playwright assertions to validate page navigation and UI feedback messages.
Week 11 **[Feb 09 to Feb 13]**	● Extended Playwright automation to cover employee creation, onboarding and offboarding. ● Tested the HRM system using different user roles including Admin, Manager, HR and Employee. ● Verified that role-based access control was correctly

	enforced, ensuring that unauthorized users were unable to access restricted features such as payroll management and employee records. ● Performed manual regression testing on HRM after new build was deployed to staging.
Week 12 **[Feb 16 to Feb 20]**	● Performed final regression testing across all three projects after last round of fixes. ● Ran full Playwright automation suite for HRM and reviewed final pass/fail report. ● Prepared consolidated defect summary for all three projects total bugs found, resolved, and deferred.
Week 13 **[Feb 23 to Feb 27]**	● Finalized and submitted all testing documentation test cases, bug reports, and execution summaries. ● Handed over Postman collections, Playwright scripts to the QA team. ● Completed internship report covering all three projects, tools, challenges, and key learnings. ● Received final performance evaluation and written feedback from mentor. ● Formally concluded internship on 27 February 2026 with certificate of completion.

3.3 Description of Projects Involved During Internship

Durimg my internship at Alpinist Studios, I was involoved in testing multiple software systems developed and maintained by the company. My work mainly involved manual testing, test case preparation, defect reporting, retesting, regression testing and API validation to support stable releases and a consistent yser experience.

3.3.1 HLDC Website

HLDC Nepal is a Nepali organization dedicated to human development and community-focused programs, likely working in areas such as livelihood support, social empowerment, and community development initiatives, with its website serving as a digital platform to present its services, programs, and mission to the public. The project included multiple testing activities which involved testing user interface components across all webpages by using Figma design references. The website functionality assessment took place on three different devices which included desktop and tablet and mobile devices through browser DevTools. The testing process covered all aspects of blog pagination and post loading and category filtering functions. The testing process included cross-browser compatibility testing which used Chrome and Firefox and Edge browsers to identify bugs that were documented in Google Sheets with accompanying screenshots.

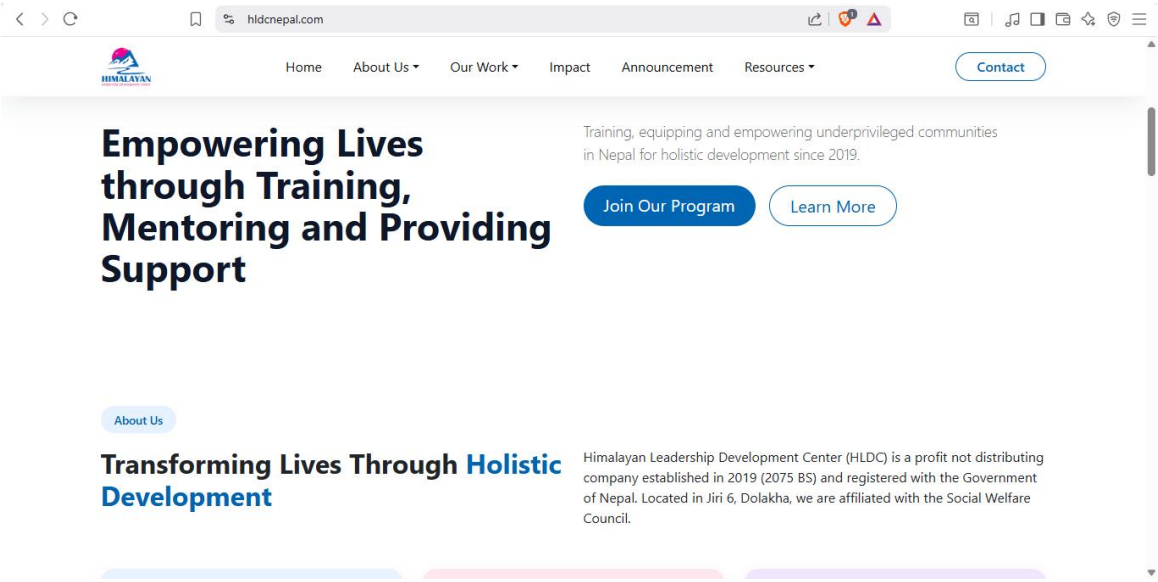


Figure 3: Homepage of HLDC Nepal website

3.3.2 SeekNFind Mobile Application

SeekNFind is a mobile application that enables users to find nearby churches while providing students and administrators and church members to manage their church memberships and create and maintain events and user profiles. For this, manual tests were conducted on the application to verify its Registration, Login, and Profile modules through test cases which covered both valid user scenarios and invalid user scenarios that included OTP verification and profile image upload. The testing team used Postman to

perform API testing because they imported Swagger documentation which allowed them to test response structures and error handling capabilities while discovering one software defect.

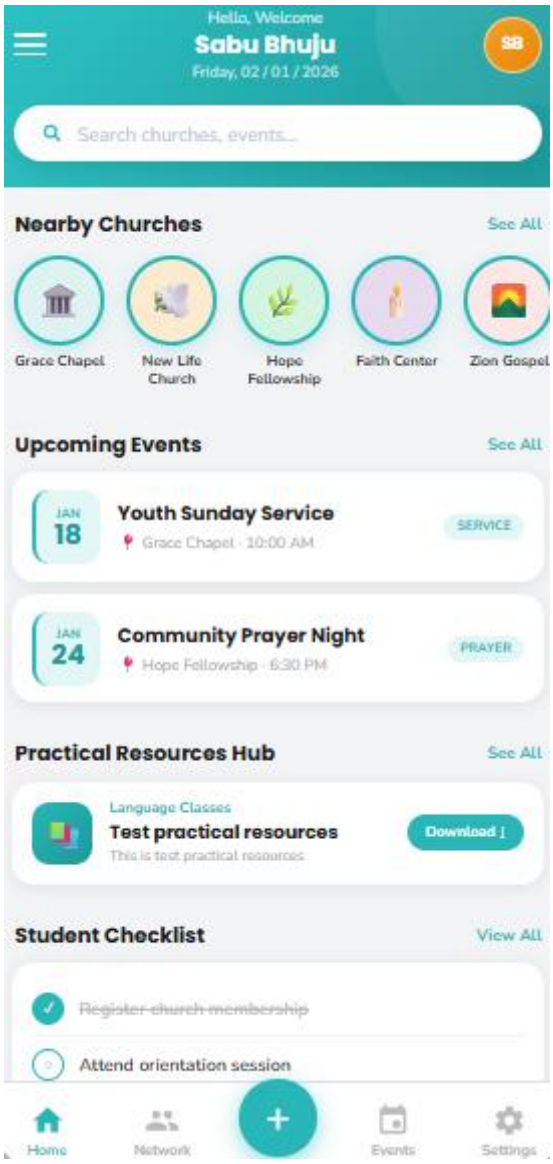


Figure 4: Homepage of SeekNFind Mobile App

3.3.3 Human Resource Management System

The HRM System functions as an online human resource management system which developed through internship work between two teams that created API functions while designing the React user interface. The project tested all main system components which included employee forms together with leave request systems and attendance tracking features. Postman was used for API testing to test employee CRUD operations while

access control functions were checked by using various user tokens during testing. The team learned Playwright automation and created automated test scripts with JavaScript for testing login employee creation employee editing employee deletion and employee onboarding/offboarding processes which they uploaded to the repository. The team performed regression testing after they deployed the system to the staging environment to verify that all existing system functions worked properly.

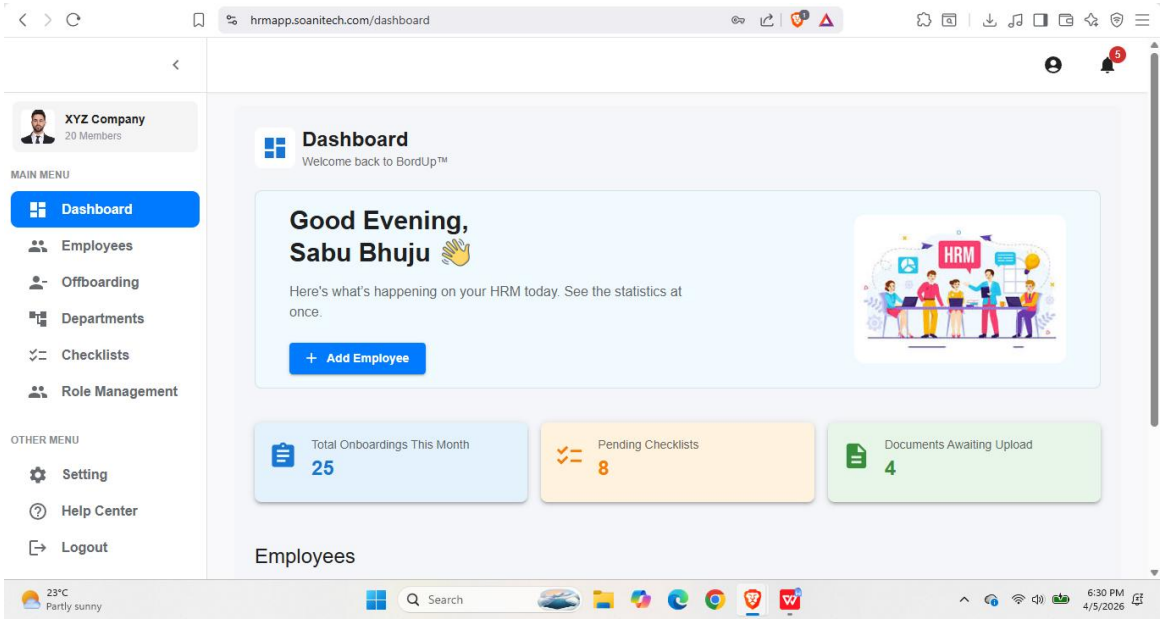


Figure 5: Homepage of HRM System

3.4 Description of the Tools

During the internship, I worked with several tools that supported testing, documentation, and team collaboration:

1. **Google sheet:** Google Sheets was used to document and track test cases, log bugs with steps to reproduce, expected and actual results, severity levels, and annotated screenshots, providing a structured and shareable format for reporting testing progress to the team.

2. **Slack**: Slack was as the primary tool for daily communication which enabled to discuss issues and clarify requirements while working with developers and project managers.

3. Jira: The team used Jira to manage their projects, track their tasks, and handle their bug issues. The Agile Test plugin in Jira was utilized to create test cases which were linked to user stories while their progress was monitored and defects were reported.

4. **Postman**: The Postman application served as the primary tool for testing APIs. The HRM System and SeekNFind mobile application were tested at all their endpoints by executing HTTP method tests and verifying response status codes and evaluating the accuracy of JSON response data.

5. **Visual Studio Code**: Visual Studio Code was used as the primary code editor for writing and managing Playwright automation scripts in JavaScript, utilizing its built-in terminal, extensions, and version control integration to edit, debug, and commit test scripts to the repository.

6. **Playwright**: Playwright was used for automated end-to-end testing of the HRM System, where JavaScript test scripts were developed covering the Login module, employee management workflows, and onboarding/offboarding processes, using assertions to validate page navigation and UI feedback messages.

3.5 Description of the Tasks and Activities Performed

The internship involved several technical activities related to software testing.

3.5.1 Manual Testing

During the internship, manual testing was conducted across all projects, including the HLDC website, SeekNFind mobile application, and HRM system. The testing process aimed to verify proper system operation according to user requirements while delivering an efficient user experience.

The **HLDC Nepal website** manual testing checked three areas which included design and usability and responsiveness testing. Testers evaluated multiple pages to verify correct alignment and consistent design and readable content and seamless navigation. The system evaluation tested its performance with various screen dimensions and multiple device types. The team identified layout misalignment and inconsistent fonts and broken links and improper spacing issues which they reported to enhance the overall user experience.

HLDC_Nepal

Bug ID	Title / Summary	Issue Type	Steps to reproduce	Expected result	Actual result	URL	Attachment	Status	Severit
BR-HOME-001	Home page button "Join Our Program" is not redirecting to any page	Bug / Defect	1. Open the website home page. 2. Locate the "Join Our Program" button. 3. Click on "Join Our Program".	Clicking "Join Our Program" should redirect the user to the relevant program registration or inforamtion page.	Clicking "Join Our Program" reload same page.	http://hldcnepal.com/	Screenshot BR-HOME-001	Fail	Medium
BR-HOME-002	Home page button "Learn More" is not redirecting to any page	Bug / Defect	1. Open the website home page. 2. Locate the "Learn More" button. 3. Click on "Learn More".	Clicking "Learn More" should redirect the user to the relevant program registration or inforamtion page.	Clicking "Learn More" reload same page.	http://hldcnepal.com/	Screenshot BR-HOME-002	Fixed	Medium
BR-HOME-003	UI Mismatch: "About Us" section's sub-heading text color does not match Figma design	Bug / Defect	1. Open the website home page. 2. Scroll down to "About us" section. 3. Observe the color of sub-heading text.	The color of the sub-heading text in About Us section should match the approved color codes specified in the Figma design.	The displayed color of sub-heading text in About Us section differs from the color defined in the Figma design	http://hldcnepal.com/	Screenshot BR-HOME-003	Fixed	Medium
BR-HOME-004	UI Mismatch: "What we do" section's sub-heading text color does not match Figma design	Bug / Defect	1. Open the website home page. 2. Scroll down to "What we do" section. 3. Observe the color of sub-heading text.	The color of the sub-heading text in What we do section should match the approved color codes specified in the Figma	As expected	http://hldcnepal.com/		Pass	Medium

Figure 6: Manual Testing for HLDC Nepal website

Manual testing of the **SeekNFind mobile application** tested its main functions and user interactions, which included user registration and login and search and navigation functions. The app was tested under scenarios like network interruptions and incorrect inputs to ensure proper error handling and stability. The development team received documentation of functional problems and input validation issues and user flow problems which were reported by testers.

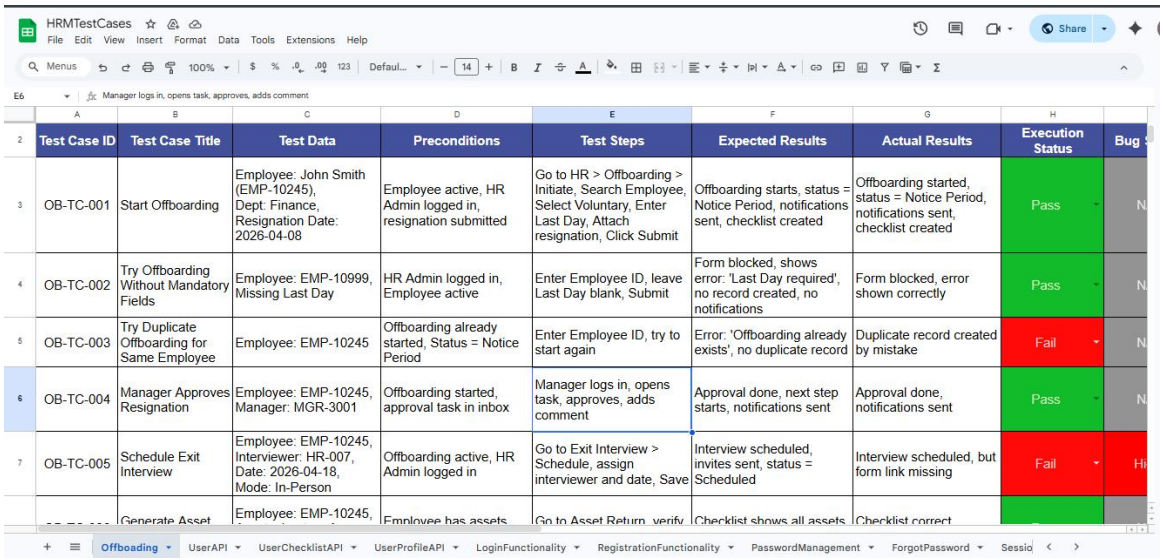
SeekNFind App Testing

Test Case ID	Test Case Title	Test Data	Preconditions	Test Steps	Expected Results	Actual Results	Execution Sta
Reg-001	Register a new user with valid email	Email: sabbubhuju@gmail.com	App must be installed and internet available	1. Open Seek&Find app. 2. Navigate to Sign Up. 3. Enter valid email. 4. Tap Continue with email.	Registration successful and send OTP to provided email for verification	Registration successful; redirected to OTP verification page.	Pass
Reg-002	Register a new user with Google	Google account exist	App installed; Internet available; User not registered	1. Tap Google login 2. Complete OAuth authentication	Google OAuth page must be open	No action triggered; screen remains unchanged	Fail
Reg-003	Register a new user with Facebook	Facebook account exist	App installed; Internet available; User not registered	1. Tap Facebook login 2. Authorize app	Facebook authorization page must be open	No action triggered; screen remains unchanged	Fail
Reg-004	Register with invalid email format	Email: sabubhuju.gmail.com	App must be installed and internet available	1. Open Seek&Find app. 2. Navigate to Sign Up. 3. Enter invalid email. 4. Tap Continue with email.	Validation error message shown	Invalid email error displayed	Pass
Reg-005	Register with already registered email	Email: sabbubhuju@gmail.com	Email already registered in the system	1. Open Seek&Find app. 2. Navigate to Sign Up. 3. Enter registered email. 4. Tap Continue with email.	Error message displayed: "Email already registered"	As expected	Pass
Reg-006	Verify email with valid OTP	OTP: ABCDEF	Valid email	1. Enter valid OTP received through email. 2. Tap verify	OTP verified; redirect to personal info form	As expected	Pass
Reg-007	Verify email with invalid OTP	OTP: 123456	Valid email	1. Enter wrong OTP . 2. Tap verify	Error message: Invalid OTP	As expected	Pass
		Account type: Student First Name: Sabu Last Name: Bhuju		1. Open app			

Figure 7: Manual Testing for SeekNFind Mobile App

Manual testing of the **HRM system** focused on employee management, onboarding/offboarding, and login functionality. Test cases were executed to verify correct data entry, storage, and retrieval. Functional testing ensured all features worked as

intended, while regression testing confirmed that bug fixes did not affect existing functionalities. Edge cases and boundary conditions were also tested to identify potential defects and improve system reliability.



Test Case ID	Test Case Title	Test Data	Preconditions	Test Steps	Expected Results	Actual Results	Execution Status	Bug
OB-TC-001	Start Offboarding	Employee: John Smith (EMP-10245), Dept: Finance, Resignation Date: 2026-04-08	Employee active, HR Admin logged in, resignation submitted	Go to HR > Offboarding > Initiate, Search Employee, Select Voluntary, Enter Last Day, Attach resignation, Click Submit	Offboarding starts, status = Notice Period, notifications sent, checklist created	Offboarding started, status = Notice Period, notifications sent, checklist created	Pass	N
OB-TC-002	Try Offboarding Without Mandatory Fields	Employee: EMP-10999, Missing Last Day	HR Admin logged in, Employee active	Enter Employee ID, leave Last Day blank, Submit	Form blocked, shows error: 'Last Day required', no record created, no notifications	Form blocked, error shown correctly	Pass	N
OB-TC-003	Try Duplicate Offboarding for Same Employee	Employee: EMP-10245	Offboarding already started, Status = Notice Period	Enter Employee ID, try to start again	Error: 'Offboarding already exists', no duplicate record	Duplicate record created by mistake	Fail	N
OB-TC-004	Manager Approves Resignation	Employee: EMP-10245, Manager: MGR-3001	Offboarding started, approval task in inbox	Manager logs in, opens task, approves, adds comment	Approval done, next step starts, notifications sent	Approval done, notifications sent	Pass	N
OB-TC-005	Schedule Exit Interview	Employee: EMP-10245, Interviewer: HR-007, Date: 2026-04-18, Mode: In-Person	Offboarding active, HR Admin logged in	Go to Exit Interview > Schedule, assign interviewer and date, Save	Interview scheduled, invites sent, status = Scheduled	Interview scheduled, but form link missing	Fail	Hi
[illegible]	Generate Asset	Employee: EMP-10245, [illegible]	Employee has assets	Go to Asset Return, verify	Checklist shows all assets	Checklist correct		

Figure 8: Manual Testing for HRM System

3.5.2 Automation Testing

The HRM system testing process was conducted through automated testing which used Playwright to evaluate the system's login, employee management, password management, and profile setup components. The team developed scripts to test functional workflows and form inputs and data handling and navigation through positive and negative test cases. The testing process included automated regression tests which confirmed that updates and bug fixes did not impact the existing system functions. The team documented all discovered issues which helped to sustain system stability and reliability and operational efficiency.

For login:

```
File Edit Selection View Go ...   HRMTEST
login.spec.js
tests > login.spec.js > ...
import { expect, test } from '@playwright/test';
const BASE_URL = 'https://hrmapp.soanitech.com/';
const VALID_EMAIL = 'admin@example.com';
const VALID_PASSWORD = 'Password12#';

test.describe('Login', () => {
  test.beforeEach(async ({ page }) => {
    await page.goto(BASE_URL);
  });

  //Test 1: Valid login (positive case)
  test('User can log in with valid credentials', async ({ page }) => {
    await page.locator('input[name="email"]').fill(VALID_EMAIL);
    await page.locator('input[name="password"]').fill(VALID_PASSWORD);
    await page.getByRole('checkbox', { name: 'Remember Me' }).check();
    await page.getByRole('button', { name: 'Login' }).click();
    await expect(page).toHaveURL(/dashboard/);
    const closeButton = page.getByRole('button', { name: 'Close' });
    if (await closeButton.isVisible()) {
      await closeButton.click();
    }
  });

  //Test 2: Wrong credentials (negative case)
  test('User cannot log in with invalid credentials', async ({ page }) => {
    await page.locator('input[name="email"]').fill('wrongemail@example.com');
```

Figure 9: Test Script Login Functionality

For Employee Management:

```
employee.spec.js
tests > employee.spec.js > loginAsAdmin

test.describe('Employee Management', () => {
  test.beforeEach(async ({ page }) => {
    await loginAsAdmin(page);
    await page.getByRole('button', { name: 'Employees' }).click();
    await expect(page).toHaveURL(/employee/);
  });

  //Test 1: Add a new employee (positive case)
  test('Admin can add a new employee successfully', async ({ page }) => {
    await page.getByRole('button', { name: '+ Add Employee' }).click();
    await expect(page).toHaveURL(/employees\/add-employee/);
    await page.getByRole('textbox', { name: 'Full Name' }).fill('Employee 1');
    await page.getByRole('textbox', { name: 'Email' }).fill('emp1@gmail.com');
    await page.getByRole('button', { name: 'Submit' }).click();
    await expect(page.getByText('User Created succesfully..')).toBeVisible();
  });

  //Test 2: Duplicate email protection (negative case)
  test('System prevents adding an employee with a duplicate email', async ({ page }) => {
    await page.getByRole('button', { name: '+ Add Employee' }).click();
    await expect(page).toHaveURL(/employees\/add-employee/);
    await page.getByRole('textbox', { name: 'Full Name' }).fill('Employee 2');
    await page.getByRole('textbox', { name: 'Email' }).fill('emp1@gmail.com');
    await page.getByRole('button', { name: 'Submit' }).click();
    await expect(page.getByText('The email has already been taken')).toBeVisible();
```

Figure 10: Test Script to Add New Employee

For Profile Management:

```
ProfileManagement.spec.js
tests > ProfileManagement.spec.js > loginUser
29  test.describe('Profile Management', () => {
47    test('User can update personal information successfully', async ({ page }) => {
48      await navigateToProfile(page);
49
50      await expect(page.getByText(/Personal/i)).toBeVisible();
51
52      const editBtn = page.getByRole('button', { name: 'Edit' }).first();
53      await expect(editBtn).toBeVisible();
54      await editBtn.click();
55
56      await expect(page).toHaveURL(/profile\/edit\/personal-information/);
57
58      await page.getByRole('textbox', { name: 'Full Name' }).fill('Sabu Bhuju');
59      await page.locator('input[name="dateOfBirth"]').fill('2001-11-02');
60      await page.getByRole('textbox', { name: 'Phone' }).fill('9832165478');
61      await page.getByRole('radio', { name: 'FEMALE' }).check();
62      await page.getByRole('textbox', { name: 'Address' }).fill('Bhaktapur');
63
64      await page.getByRole('textbox', { name: 'Emergency Name' }).fill('Emi');
65      await page.getByRole('textbox', { name: 'Emergency Contact' }).fill('9875461230');
66
67      await page.getByRole('button', { name: 'Submit' }).click();
68
69      await expect(page.getByText('Personal info saved')).toBeVisible();
70    });
```

Figure 11: Test Script Profile Setup

3.4.3 API Testing

The **HRM system** and **SeekNFind application** APIs were tested using Postman as the primary evaluation tool. The testing team executed requests using HTTP methods such as GET, POST, PUT, DELETE and PATCH to validate key functionalities. The HRM system required testing of employee data retrieval APIs and onboarding/offboarding APIs and login APIs and profile update APIs. The SeekNFind system required validation of user registration APIs and search APIs and booking operation APIs. The team conducted complete response evaluations which included status code examination and response format verification and data integrity testing.

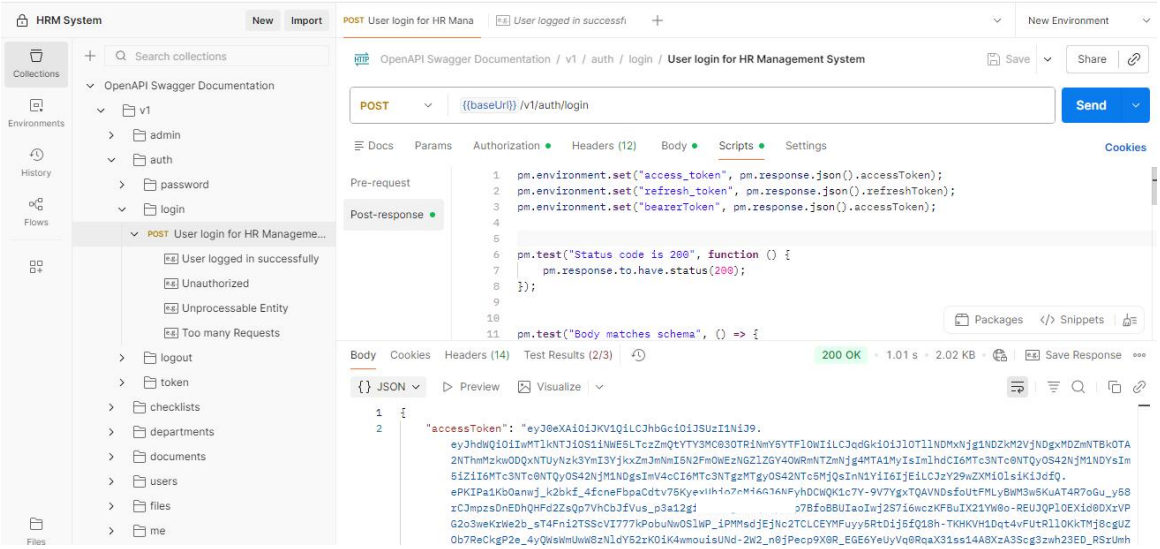


Figure 12: API Testing of HRMS User Login

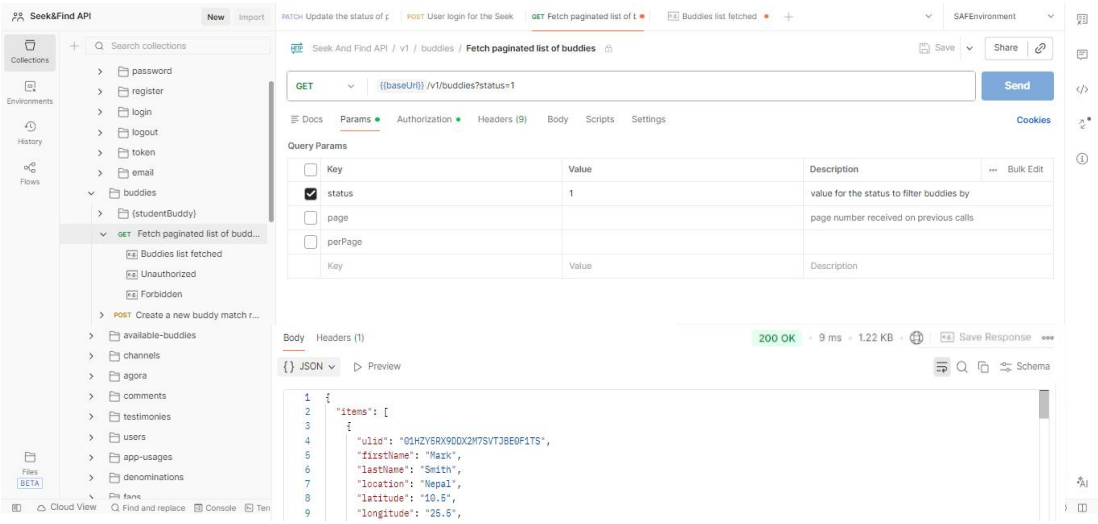


Figure 13: Buddy Match API Test of SeekNFind App

3.4.4 Positive Testing

The application shows correct functionality when valid data and expected inputs are used for testing. The process assumes users will follow the correct sequence while using the system according to its intended design. The application shows correct performance through positive testing because it meets all expected results during normal operation. The test fails when positive testing encounters an unexpected error or displays improper system behavior.

```
Password.specjs
tests > Password.specjs > test.describe('Change Password') callback > test('[Positive] User can change password with correct old password and valid new pa
25  test.describe('Change Password', () => {
43
44    //Test 2: Successfully change password with valid inputs
45    test('[Positive] User can change password with correct old password and valid new password', async ({ page }) => {
46      await page.getByRole('textbox', { name: 'Old Password' }).fill('Password123#');
47      await page.locator('button').nth(3).click();
48
49      await page.getByRole('textbox', { name: 'New Password' }).fill('Password12#');
50      await page.locator('button').nth(4).click();
51
52      await page.getByRole('textbox', { name: 'Confirm Password' }).fill('Password12#');
53      await page.locator('button').nth(5).click();
54
55      await page.getByRole('button', { name: 'Submit' }).click();
56
57      // Verify success message (typo "Succesfully" matches the actual app message)
58      await expect(page.getByText('Password Changed Succesfully')).toBeVisible();
59    });
60
61
```

Figure 14: Positive Testing to Change User Password

3.4.5 Negative Testing

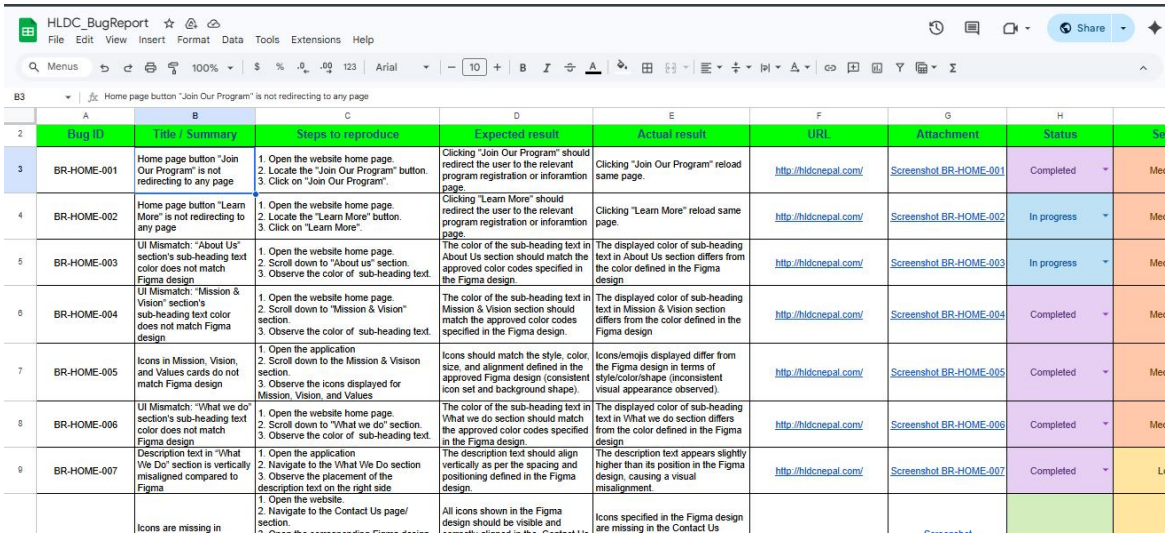
The application needs to show proper handling of invalid data and unexpected input because negative testing verifies this requirement. The test needs to handle invalid user data submission which includes incomplete information and out-of-bounds data because the system must create correct error messages and validation feedback. The application demonstrates its ability to handle all situations that exceed its expected operational limits through negative testing procedures. The test results in a failure when the application does not manage special inputs correctly or when it exhibits unpredictable behavior.

```
Password.specjs
tests > Password.specjs > test.describe('Change Password') callback > test('[Positive] User can change password with correct old password and valid new pass
25  test.describe('Change Password', () => {
61
62    // — Test 3: Wrong old password —
63    test('[Negative] System rejects password change when old password is incorrect', async ({ page }) => {
64      await page.getByRole('textbox', { name: 'Old Password' }).fill('WrongPassword99#');
65      await page.getByRole('textbox', { name: 'New Password' }).fill('Password123#');
66      await page.getByRole('textbox', { name: 'Confirm Password' }).fill('Password123#');
67
68      await page.getByRole('button', { name: 'Submit' }).click();
69
70      await expect(page.getByText('Current password is incorrect')).toBeVisible();
71    });
72
73    // — Test 4: New password same as old password —
74    test('[Negative] System rejects password change when new password is same as old password', async ({ page }) => {
75      await page.getByRole('textbox', { name: 'Old Password' }).fill('Password123#');
76      await page.getByRole('textbox', { name: 'New Password' }).fill('Password123#');
77      await page.getByRole('textbox', { name: 'Confirm Password' }).fill('Password123#');
78
79      await page.getByRole('button', { name: 'Submit' }).click();
80
81      await expect(page.getByText('The new password field and old password must be different.')).toBeVisible();
82    });
```

Figure 15: Negative Tesing to Change User Password

3.4.6 UI/UX Testing

The HLDC Nepal website underwent UI/UX testing to create an interface which provides users with an easy-to-navigate and consistent experience. The team used Figma reference design to assess page layout and component alignment and color and font and icon and complete design elements. The team tested navigation links and interactive elements which included buttons and forms and menus to verify that users could navigate the system without problems. Developers received design inconsistencies which the team documented through screenshots in Google Sheets for resolution purposes.



Bug ID	Title / Summary	Steps to reproduce	Expected result	Actual result	URL	Attachment	Status	Se
BR-HOME-001	Home page button "Join Our Program" is not redirecting to any page	1. Open the website home page. 2. Locate the "Join Our Program" button. 3. Click on "Join Our Program".	Clicking "Join Our Program" should redirect the user to the relevant program registration or inforamtion page.	Clicking "Join Our Program" reload same page.	http://hldcnepal.com/	Screenshot BR-HOME-001	Completed	Me
BR-HOME-002	Home page button "Learn More" is not redirecting to any page	1. Open the website home page. 2. Locate the "Learn More" button. 3. Click on "Learn More".	Clicking "Learn More" should redirect the user to the relevant program registration or inforamtion page.	Clicking "Learn More" reload same page.	http://hldcnepal.com/	Screenshot BR-HOME-002	In progress	Me
BR-HOME-003	UI Mismatch: "About Us" section's sub-heading text color does not match Figma design	1. Open the website home page. 2. Scroll down to "About us" section. 3. Observe the color of sub-heading text.	The color of the sub-heading text in About Us section should match the approved color codes specified in the Figma design.	The displayed color of sub-heading text in About Us section differs from the color defined in the Figma design	http://hldcnepal.com/	Screenshot BR-HOME-003	In progress	Me
BR-HOME-004	UI Mismatch: "Mission & Vision" section's sub-heading text color does not match Figma design	1. Open the website home page. 2. Scroll down to "Mission & Vision" section. 3. Observe the color of sub-heading text.	The color of the sub-heading text in Mission & Vision section should match the approved color codes specified in the Figma design.	The displayed color of sub-heading text in Mission & Vision section differs from the color defined in the Figma design	http://hldcnepal.com/	Screenshot BR-HOME-004	Completed	Me
BR-HOME-005	Icons in Mission, Vision, and Values cards do not match Figma design	1. Open the application 2. Scroll down to the Mission & Visison section. 3. Observe the icons displayed for Mission, Vision, and Values	Icons should match the style, color, size, and alignment defined in the approved Figma design (consistent icon set and background shape).	Icons/emojis displayed differ from the Figma design in terms of style/color/shape (inconsistent visual appearance observed).	http://hldcnepal.com/	Screenshot BR-HOME-005	Completed	Me
BR-HOME-006	UI Mismatch: "What we do" section's sub-heading text color does not match Figma design	1. Open the website home page. 2. Scroll down to "What we do" section. 3. Observe the color of sub-heading text.	The color of the sub-heading text in What we do section should match the approved color codes specified in the Figma design.	The displayed color of sub-heading text in What we do section differs from the color defined in the Figma design	http://hldcnepal.com/	Screenshot BR-HOME-006	Completed	Me
BR-HOME-007	Description text in "What We Do" section is vertically misaligned compared to Figma	1. Open the application 2. Navigate to the What We Do section 3. Observe the placement of the description text on the right side	The description text should align vertically as per the spacing and positioning defined in the Figma design.	The description text appears slightly higher than its position in the Figma design, causing a visual misalignment.	http://hldcnepal.com/	Screenshot BR-HOME-007	Completed	L
	Icons are missing in	1. Open the website. 2. Navigate to the Contact Us page/ section.	All icons shown in the Figma design should be visible and	Icons specified in the Figma design are missing in the Contact Us				

Figure 16: UI Test Cases for HLDC Nepal website

CHAPTER 4: CONCLUSION AND LEARNING OUTCOMES

4.1 Conclusion

The internship at **Alpinist Studios** as a **Quality Assurance (QA) and Quality Control (QC) Intern** was found to be an immensely beneficial and practical educational experience. The internship provided training in software testing methods which included manual testing API testing and testing of frontend and mobile applications together with defect management.

Theoretical knowledge of software testing and the Software Testing Life Cycle (STLC) was applied to execute real-world projects through work on the HRM System the SeekNFind mobile application and the HLDC website. The experience provided knowledge about how Quality Assurance teams confirm that software products achieve exceptional quality and dependability standards.

The program provided essential knowledge about how teams work together and how developers and project managers communicate and how they make decisions regarding software quality and feature development. The program helped to develop technical skills which enabled better problem-solving and information analysis abilities.

4.2 Learning Outcome

The following learning outcomes were achieved during the internship:

1. Acquired practical knowledge about software testing essential elements, testing fundamental principles, and testing methods which included manual testing, functional testing, and non-functional testing.
2. Acquired abilities to write comprehensive test cases and test scenarios and test documentation through Google Docs.
3. Conducted API testing through Postman while checking HTTP methods and response codes and data integrity.

4. Conducted usability testing and functional testing and interface responsiveness testing of web applications developed with React and mobile applications developed with Flutter.
5. Learned how to identify defects and report them and track their progress and resolve them while determining their severity and priority levels.
6. Developed professional soft skills through teamwork and communication and problem-solving and time management development within a real project environment.
7. Studied basic automation testing framework concepts through Playwright testing and performance testing with JMeter.
8. Discovered that QA testing serves as a vital function which must ensure software products achieve reliability and scalability and usability throughout their usage in actual business settings.

The internship program established connections between academic study and professional work which equipped me with essential skills needed for my future career in software quality assurance and testing.

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APPENDIX

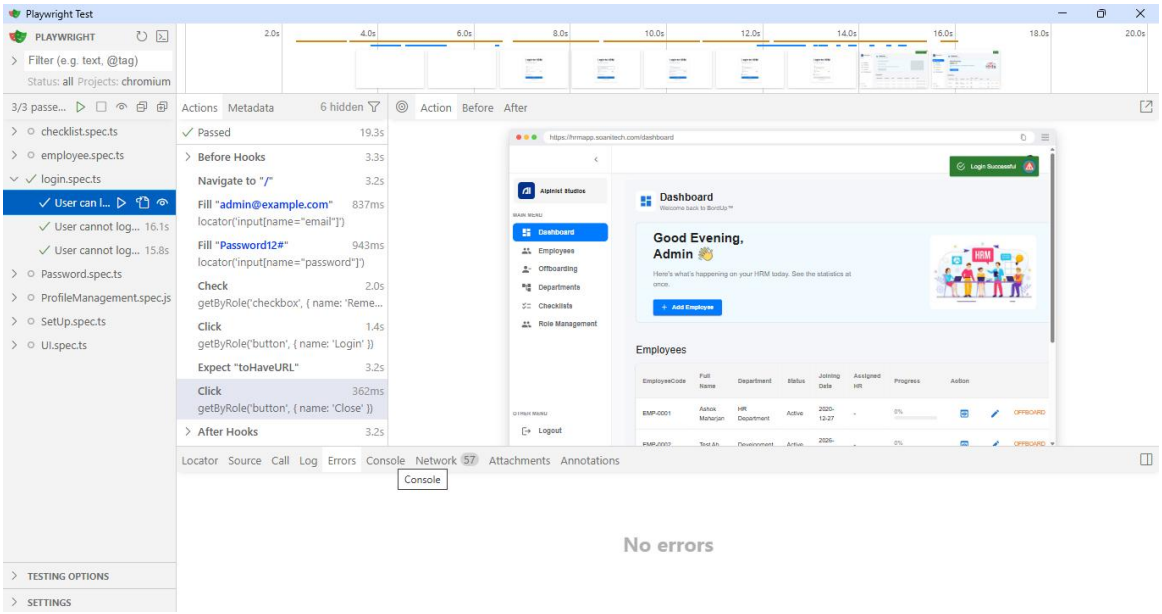

Figure 17: Playwright Test UI after successful Login

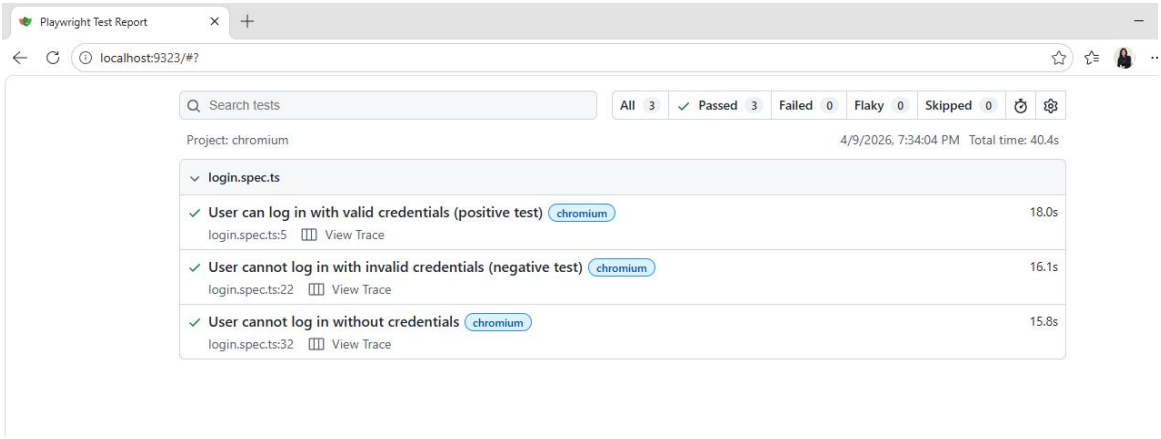

Figure 18: Playwright Report of Login Functionality

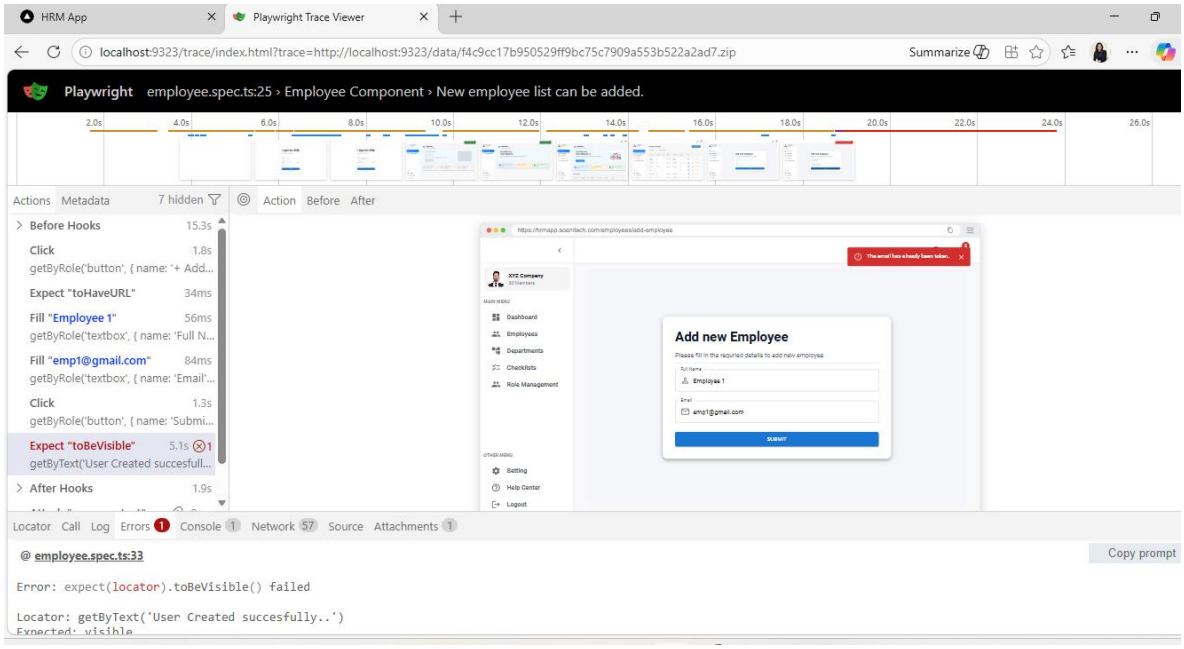

Figure 19: Playwright Test UI for Adding Duplicate Employee

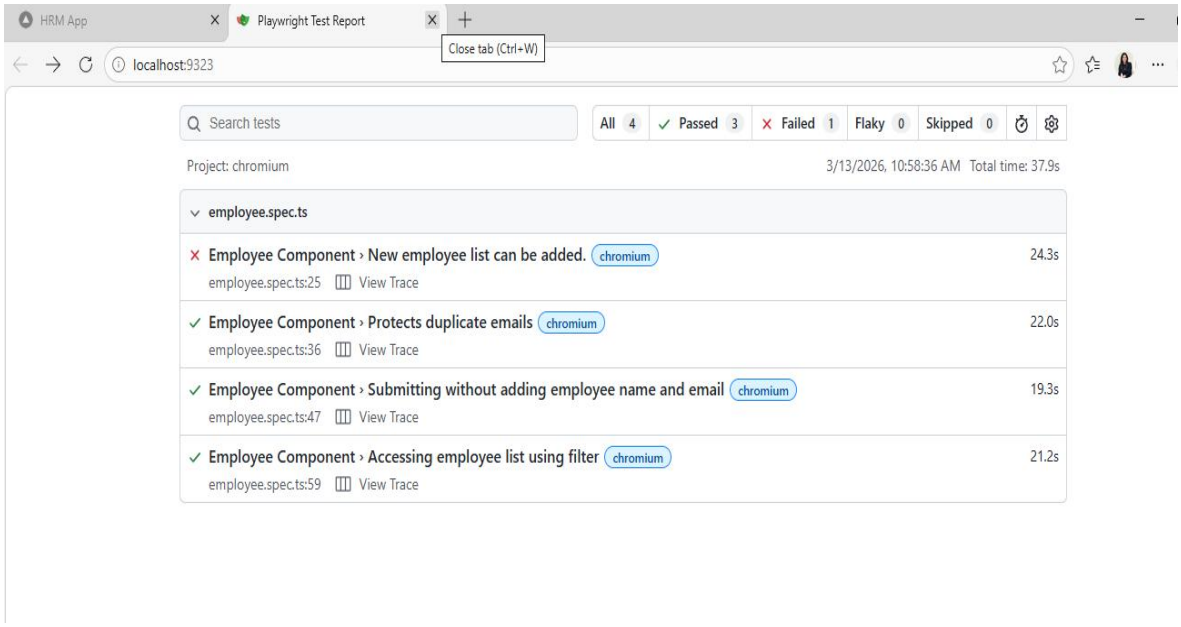

Figure 20: Playwright Report for Adding New Employee